

**ADMINISTRATIVE INFORMATION****FMI No: 25307****FMI Title: LightSpeed Plus MDAS De-Population**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS-AM: Code AM	Issue Date:
☒ Mandatory	GEMS-E: Code E	Close Date:
☐ Mandatory on Request	GEMS-A: Code A	

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☒ Batch with	<u>FMI # 25298, 25299, 25300</u> To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:** 2281177 LightSpeed Ultra Gantry Assembly**FMI Instructions Part No.:** 2331992-100**Do NOT leave unsecured on site.**



Completion Certification Report

Mail To

GEMS - Americas

GE Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - E / ESC
FMI Admin. 322
BP34
F 78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25307 FMI Title: LightSpeed Plus MDAS De-Population

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code:

- 1.) Modification Complete or previously modified.
- 2.) FMI not required per FMI instruction.
- 3.) Not modified. Unit in storage. FMI kit returned to stock.
- 4.) Modification refused or equipment altered. Customer signature and title required.
- 5.) Unit location unknown or unit not in area. Product Locator notified.

**Customer Refusal or Equipment Altered
- Modification Pending.**

Installer
Comments

Effectivity

The following CT Equipment is affected:

2281177 LightSpeed Ultra Gantry Assembly

Effectivity of Gantry Serial Numbers

272140CN3	273735CN1	274017CN3	274279CN9	274331CN8	276194CN8	276689CN7
276738CN2	276851CN3	276859CN6	276867CN9	276920CN6	276922CN2	277105CN3
277150CN9	277220CN0	277221CN8	277759CN7	277961CN9	279103CN6	281421CN8
282187CN4	282316CN9	282336CN8	282640CN2	282652CN7	282653CN5	282934CN9
282935CN5	282937CN2	283007CN3	283008CN1	283398CN6	283398CN6	

Purpose

The LightSpeed Ultra/Plus system is designed with the intention of supporting both a 4 Slice and 8 Slice MDAS configuration. All systems built in 2002 were fully populated 8 Slice MDAS systems with the intention of being upgraded from the 4 Slice M3 software to 8 Slice software. Some sites delivered in 2002 have not purchased the 8 Slice option and therefore will not be upgraded to 8 Slice software.

At the introduction of the LightSpeed Plus/Ultra M4 Software, a 4 Slice scanner must only contain 24 DAS converter cards. This was done to minimize the cost of the 4 Slice scanner to the customer. With this configuration the 4 Slice FET modes are different than with a fully populated (48 card) MDAS. For this reason, any LightSpeed Ultra/Plus scanner without the 8 Slice option enabled must have the 24 extra DAS converter cards removed. Failure to configure the hardware correctly will be sensed by the system resulting in configuration mismatch errors which will prohibit scanning.

Note: **PRIOR TO PERFORMING THIS FMI, CONFIRM WITH YOUR LOCAL SALES TEAM THAT AN ORDER FOR 8 SLICE OR THE 8 SLICE FLEX TRIAL IS NOT IN EFFECT FOR THE CUSTOMER.**

Prerequisite FMIs

- None

Related FMIs

- FMI 25300
- FMI 25298
- FMI 25299

Furnished Materials

Qty	Description	Part #
1	FMI Document	2331992-100
2	Spacer, MDAS Weight w/holes (Aluminum Tubes)	2288322-2
4	Weight, 4 Slice MDAS Balance	2328617
4	Screw Hexagon Socket 4 mm X 50 mm	46-328717P77
4	Washer Lock - Spring 4.1 mm 7.6 mm	46-328432P2
1	Loctite 242, 0.5cc Tube	46-170686P1
24	Pouch, Anti-static, 8" x 12" 3 layer with zip	2335502
24	Faraday ESD Bag, 10" x 12" Zip Top	2335503
1 box	nitrile gloves large	2207303-6
1 box	nitrile gloves extra large	2207303-7
1	Helios 2 SmartSpeed Option	2284205
1	ThinTwin Helical Option	2305988
1	Helical Tilt Software Option	2305990
1	Recon Plus Software Option	2305991
2	Label "FMI25307 Completed"	

Required Tools

Qty	Description	Part #
1	Field Torque Wrench Kit	46-268445G1
1	Field ESD Kit	2220482
1	Screwdriver Phillips # 2	Field Tool
1	8 mm hex key	Field Tool
1	3 mm hex key for 1/4 inch socket drive	Field Tool
1	19 mm (3/4 inch) crowfoot extension (Found in Balance Weight Kit shipped with system)	2309583
1	1/4 inch socket drive set	Field Tool
1	1/4 inch extensions (12 inch total length)	Field Tool

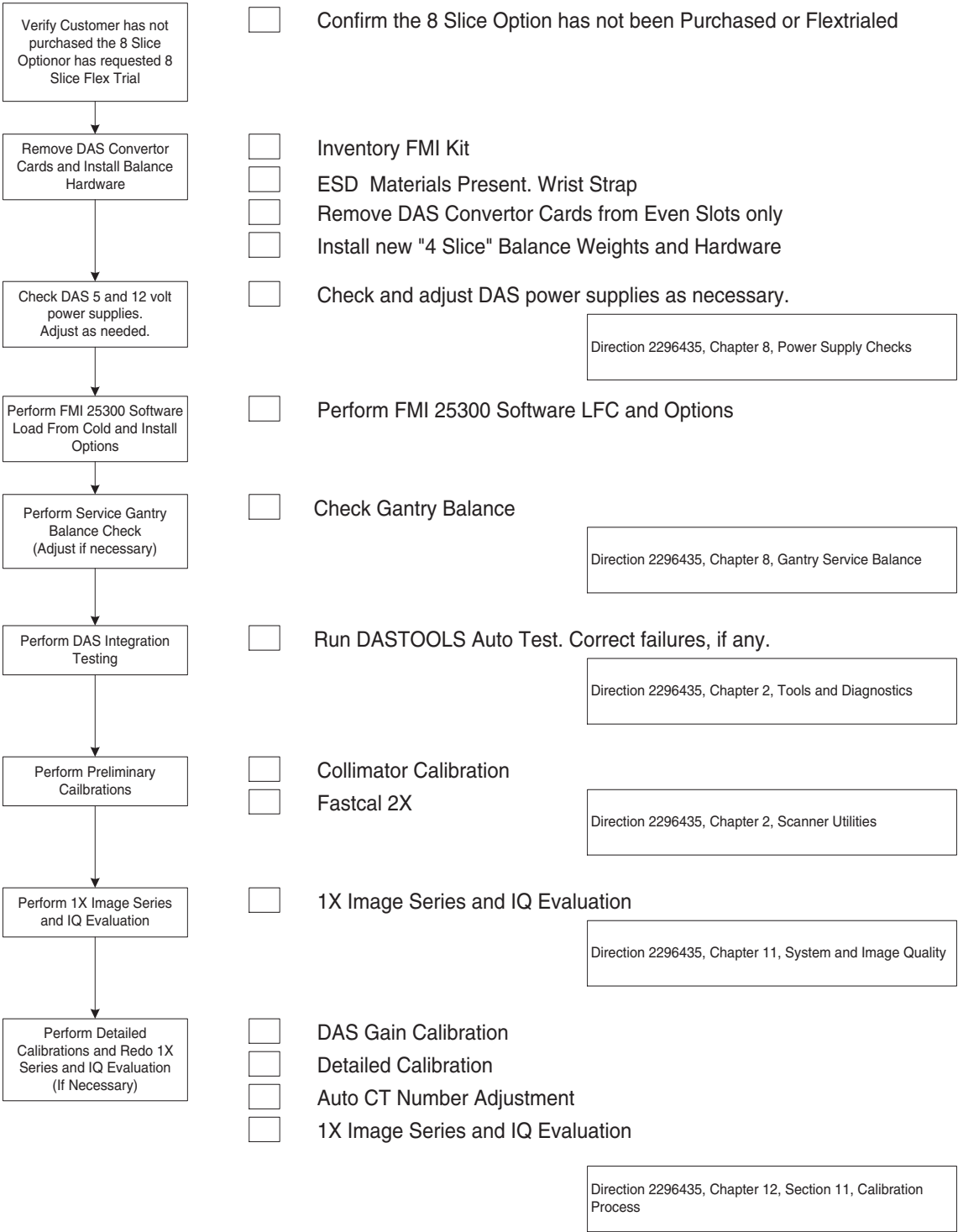
People Power

“16” hours labor

Time	Task
1.0 hour	DAS De-population and Balance Weight Installation
4.0 hours	Software LFC and Options Installation FMI 25300
0.5 hours	Service Gantry Balance Procedure
1.5 hours	DAS Functional Checks
1.0 hours	Preliminary Calibration
2.0 hours	1X Series and Image Analysis
4.0 hours	Detailed Calibration (If needed)
2.0 hours	1X Series and Image Analysis (If needed)

Procedure

1 Procedural Steering Diagram



2 DAS De-Population and Gantry Balance

2.1 DAS Convertor Card Removal

- 1.) Move the table cradle to the home position, and lower the table to its lowest elevation.
- 2.) Remove right side cover.
- 3.) Turn off all 3 switches (Axial Drive, **HVDC**, 120VAC) on the **STC** backplane.
- 4.) Remove gantry top, side, and front covers.
- 5.) Rotate gantry so that the **DAS** assembly is at the 12:00 position.



WARNING



POTENTIAL FOR PERSONAL INJURY. USE THE ROTATIONAL LOCK TO PREVENT UNCOMMANDED MOVEMENT OF THE GANTRY.

- 6.) Lock the gantry in position, using the rotational lock.



NOTICE



Follow ALL Electrostatic Discharge procedures. Always use ESD Wrist Strap and grounding cords. The use of a ground Monitor is suggested.

- 7.) Put on wrist strap and use **ESD** precautions.
Jack locations for grounding the wrist strap are near the left and right chassis mounting screws.

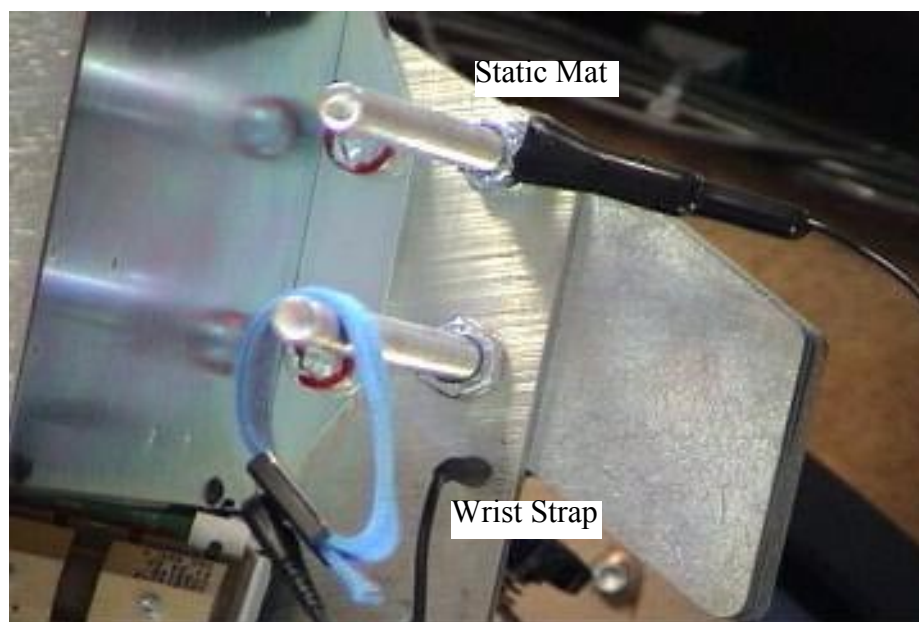


Figure 1 - ESD Ground Point Locations

- 8.) Remove DAS Chassis covers. Take care not to damage the fiber optic cable on the right chassis connected to the DCB board. [See Figure 2 on page 6.](#)
- 9.) Remove Convertor cards one at a time and place in the provided ESD bags.

Note: There are two different static bags, 1 thin and 1 with padding. Insert the Convertor card into the thin ESD bag first, then insert this into the padded ESD bag.

- a.) Start with Convertor card in slot 2, then 4, then 6 through slots 48.

Note: **MAKE SURE YOU REMOVE CARDS FROM EVEN SLOTS ONLY STARTING AT SLOT NUMBER 2.** [See Figure 3 on page 6.](#)

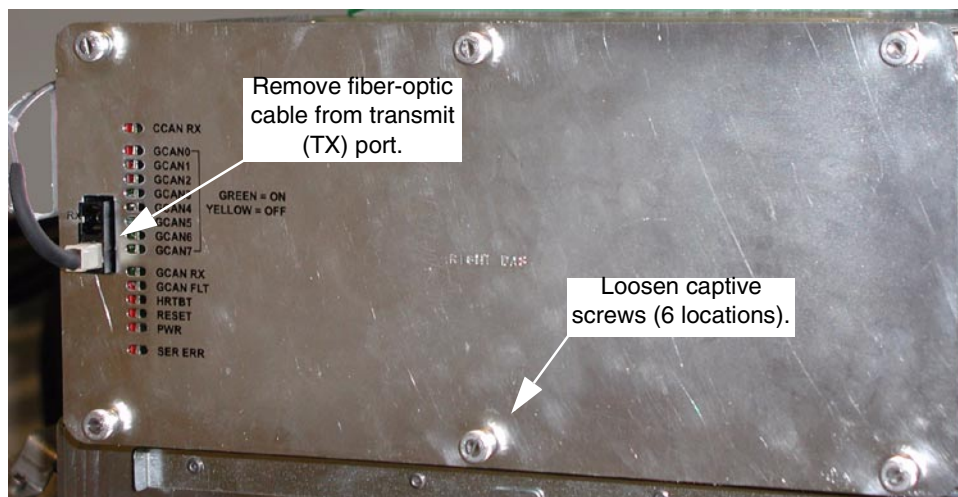


Figure 2 - DAS Chassis Cover, right chassis shown

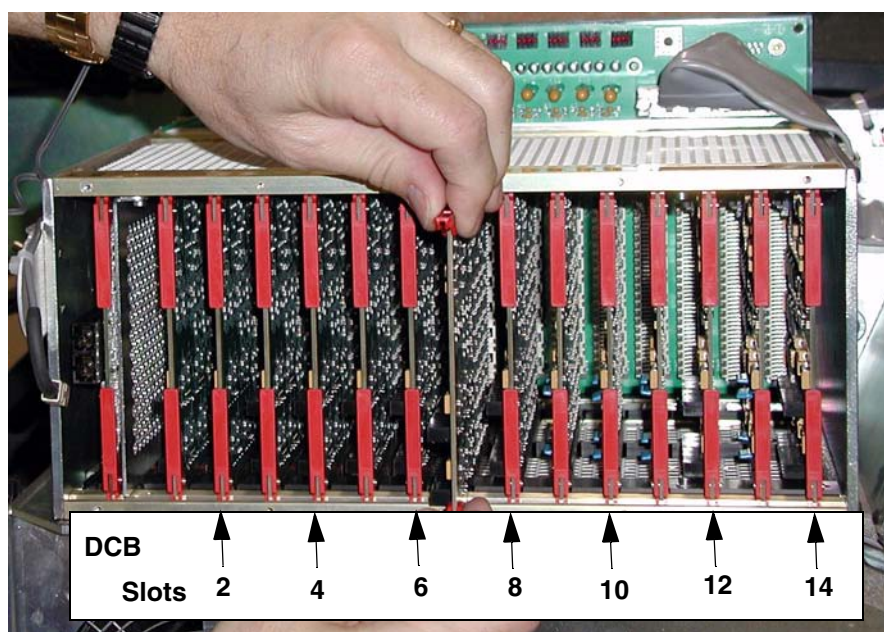


Figure 3 - Converter Card removal, Slot 7 shown

- 10.) Re-install DAS Chassis covers and connector fiber optic cable to DCB TX port.
- 11.) Install FMI Completion Label on left Gantry Support post next to MFG rating plate.
- 12.) Install FMI Completion Label on each DAS Chassis next to MFG rating plates.

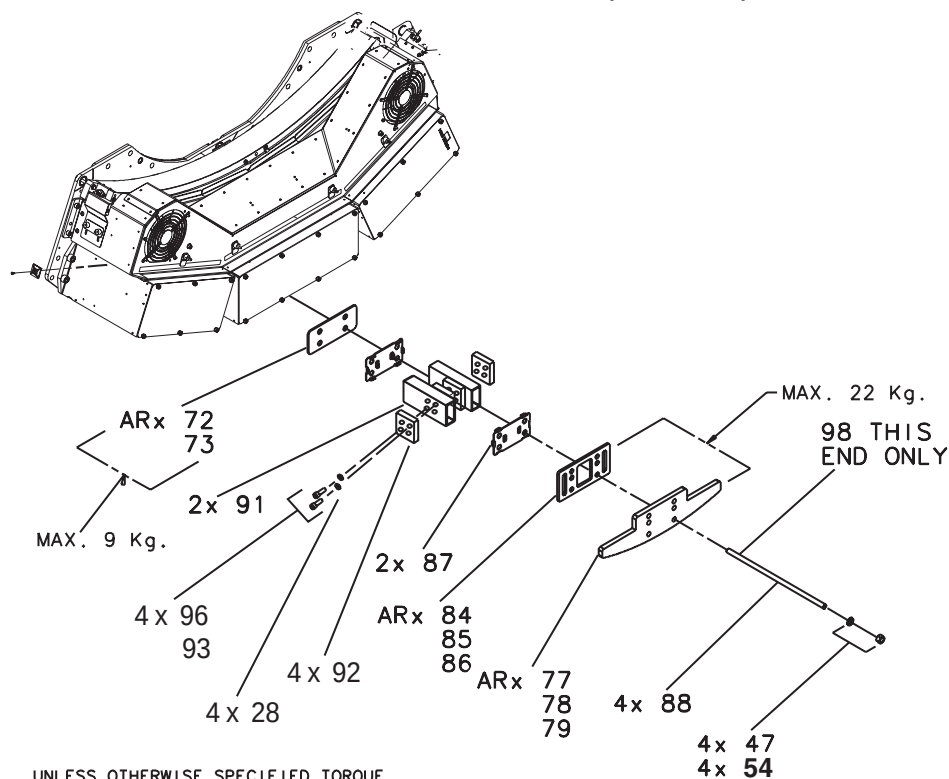
2.2 Gantry Balance Weight Installation



DANGER

FAILURE TO CORRECTLY ASSEMBLE AND TORQUE THE WEIGHT ASSEMBLY WILL RESULT IN CATASTROPHIC PROJECTILE FAILURE. RESULTS INCLUDING GANTRY HARDWARE DAMAGE AND POTENTIAL PATIENT INJURY.

2291276-2 HELIOS 3 ROTATING BASE (4 SLICE) ASM ONLY



UNLESS OTHERWISE SPECIFIED TORQUE FASTENERS USING THE FOLLOWING VALUES, HEX SOCKET HEAD CAP SCREWS ONLY VALUES ARE BASE ON ENGAGING LIKE MATERIALS OR HAVING TWO DIAMETERS OF BOLT ENGAGEMENT

FASTENER SIZE	TORQUE
M3	1 Nm
M4	2,3 Nm
M6	7,9 Nm
M10	38,4 Nm
M12	66,4 Nm
M16	160 Nm

Item 96 Screw Hexagon Socket 4MM

Item 93 Loctite 242

Item 92 Weight, MDAS 4 Slice Only Balance

Item 91 Spacer, MDAS Weight w/holes (Aluminum Tube)

Item 28 Washer, Lock Spring 4.1 MM 7.6MM Spring Steel

Item 54 Nut Hexagon 12 MM

Item 47 Washer, Lock Spring 12.2 MM 21.1 MM

Figure 4 - 4 Slice Gantry Balance Weight Assembly Diagram

2.2.1 Disassembly Procedure

- 1.) Remove and save for reuse the following items. Refer to [Figure 4](#) as a cross reference.
 - a.) Make sure you keep ALL items in order. Use the cradle to maintain arrangement.
 - Item 54 and 47 (12 mm nut and lock washers)
 - Items 77, 78, 79 (Permanent Weights)
 - Items 84, 85, 86 (Spacers)
 - Item 87 (Aluminum Tube Alignment Guide)
 - Items 91 (Aluminum Tubes)
 - Do not remove Items 72 and 73 and rear Aluminum Tube Alignment Guide.
- 2.) Discard the 2 Aluminum Tubes, item 91.

2.2.2 Assembly Procedure

- 1.) Loosely assemble the new Aluminum Tubes and 4 Slice Balance weights before you install on the gantry. Make sure the "4 Slice" weights move freely.



CAUTION

Do not tighten the weights onto the aluminum tube. You can damage the tubes by crushing them resulting additional downtime awaiting replacement parts.

- a.) Assembly order
 - I.) 4 mm Hexagon screw
 - II.) Lock washer
 - III.) 4 Slice weight
 - IV.) Aluminum Tube
 - V.) 4 Slice weight
- b.) Reference [Figure 5](#) for correct hole selections. The 4 mm hexagon screws fasten to the inner 4 Slice weight.

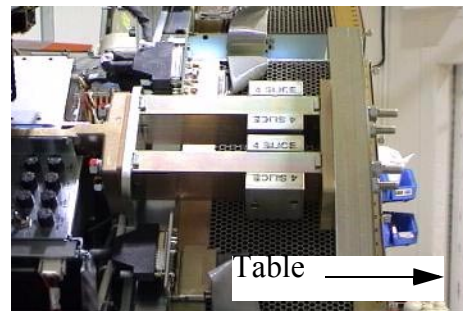
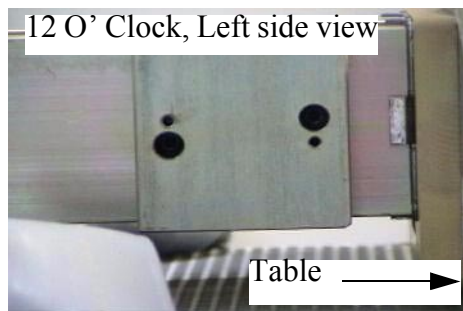


Figure 5 - 4 Slice Weight and Aluminum Tube Assembly

- 2.) Install new 4 Slice Weight assembly onto the threaded rod.
- 3.) Install item 87, Aluminum Tube Alignment Guide.



CAUTION

Make sure the tabs on the alignment guides are on the outside of the aluminum tubes both front and back.

- 4.) Install Spacer, item 84, 85, 86.
- 5.) Install Permanent Weights, items 77, 78, 79.
- 6.) Install Lock Washers and 12 mm Hex nuts, items 47, 54.
- 7.) Ensure Assembly is properly aligned.

- 8.) Torque 12 mm hex nuts to 66.4 N-m (49 lb-ft) FOOT POUNDS.
- 9.) Remove 1 at a time, the 4 mm Hexagon screws from the 4 Slice Weights.
 - a.) Add 2 drops of Loctite 242 and re-install.
 - b.) Torque to 2.3 N-m (20.4 lb-in) INCH POUNDS.
- 10.) Unlock the Gantry Rotational Lock.
- 11.) Restore Gantry power at the STC backplane. Enable Axial Drives.

Torque Reference Information

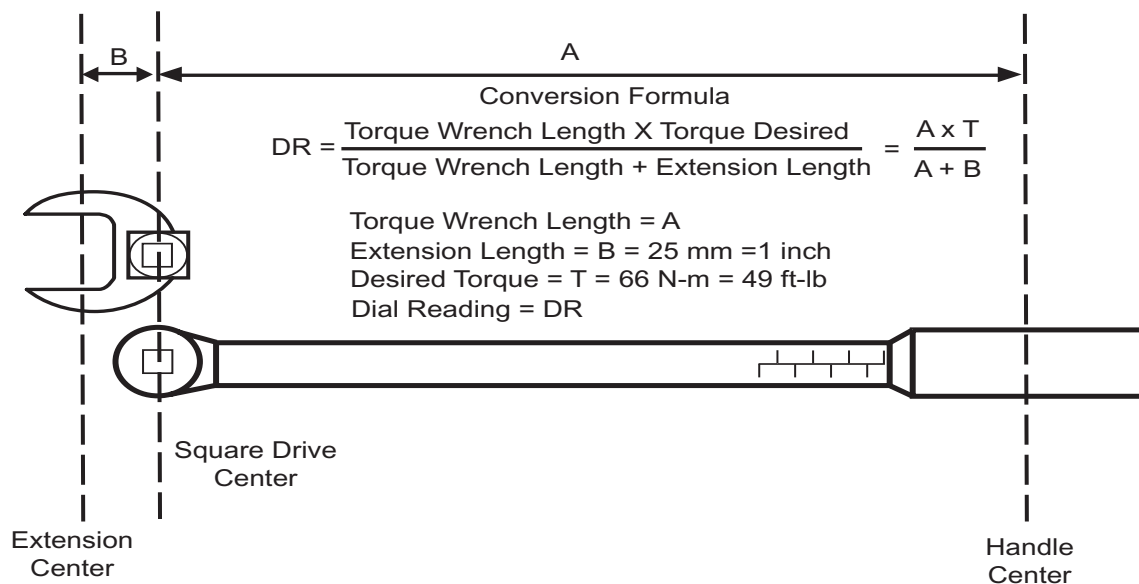


Figure 6 - Torque Wrench Usage Reference

3 DAS Power Supply Check

Since 24 boards have been removed, you need to check the DAS 5 and 12 volt power supplies. There is a strong possibility that the change in current load has resulted in voltage levels that are out of specification. Refer to the System Service Manual, Chapter 8, Power Supply Checks, DAS Power Supplies for details.

4 SW Load From Cold, FMI 25300

At this point you need to perform FMI 25300 to install the current software platform that supports the 4/8 Slice product configuration.

5 Service Gantry Balance

- 1.) Insert proprietary key.
- 2.) Click on Display monitor, **SERVICE ICON.**
- 3.) Click **CHECK SECURITY** from Desktop Tool Chest, upper right of monitor.
- 4.) Click on Display monitor, **EXAM RX ICON** and wait 10 seconds.
- 5.) Click on Display monitor, **SERVICE ICON.**

- 6.) Enter security code for Proprietary Access. Click 1, 2, 3.
- 7.) Click **REPLACEMENT PROCEDURES**
- 8.) Click **GANTRY SERVICE BALANCE**
 - Gantry should be balanced.
 - If the Balance check reports the system is out of balance then continue following instructions within the Balance Program until the gantry is balanced.

6 Functional Testing

Launch and run 1 full pass of auto test in DASTOOLS. Correct any failures that may occur. From Applications Level perform the following to launch DASTOOLS.

- 1.) Click on Service Desktop **DIAGNOSTICS** icon.
- 2.) Click **DATA ACQUISITION**.
- 3.) Click **DASTOOLS**.
- 4.) Click on **AUTO TEST** within DASTOOLS and Click **RUN**.

7 Preliminary Calibration

Perform Collimator Calibration and FASTCAL to establish a baseline for IQ checks.

- 1.) Click Left monitor **SCANNER UTILITIES** on bottom of the left monitor.
- 2.) Click **COLLIMATOR CALIBRATION** from the displayed selections.
- 3.) Click **QUIT**.
- 4.) Click **DAILY PREP**.
- 5.) Click **FASTCAL**.
- 6.) Click **FASTCAL** and run a second time.

8 1X Image Series

Reference the *LightSpeed Ultra System Service Manual - Advanced*, DIRECTION 2296435-100, REVISION 3, Book 6, Chapter 11 for details on how to perform the 1X Image Series. Below is an outline of the required scans. This Chapter also contains information on how to use Image Analysis to evaluate the 1X series images.

Phantom	kVp	mA	Slice	Time	#	# img	CT #	Noise	Unif	MTF	LCD	Ring	Band	Streak	Smudge	Clump	CA
48 Poly	120	340	4x5(4i)	1	4	16	NA	<50	±8.5	NA		*	*	*	*	*	NA
	120	200	4x5(4i)	0.8	4	16	NA	<70	±8.5	NA		*	*	*	*	*	NA
	120	400	4x1.25(4i)	2	4	16	NA	<50	±8.5	NA		*	*	*	*	*	NA
GE	120	440	4x5(2i)	0.8	4	4	NA	NA	NA	*		NA	NA	NA	NA	NA	NA
Perfor	120	130	4x5(2i)	2	4	4	NA	NA	NA	*		NA	NA	NA	NA	NA	NA
	120	40	4x5(2i)	0.8	4	4	NA	NA	NA	*		NA	NA	NA	NA	NA	NA
0.8 sec	120	440	5mm 7.5Q	HE	4	4	NA	NA	NA	*		NA	NA	NA	NA	NA	NA
GE QA	120	260	4x5(2i)	1	4	4	NA	NA	NA	*		NA	NA	NA	NA	NA	NA
	120	260	4x5(2i)	1	4	4	NA	NA	NA	NA	*	NA	NA	NA	NA	NA	NA
	120	260	4x5(2i)	1	4	8	±3.0	<0.3 3	±3.0	NA	NA	*	*	*	*	NA	*
	120	200	4x1.25(4i)	2	4	16	±3.0	NA	±3.0	NA	NA	*	*	*	*		
	120	400	4x5(4i)	2	4	16	±3.0	NA	±3.0	NA	NA	*	*	*	*	NA	*
	120	400	4x1.25(4i)	1	4	16	±3.0	NA	±3.0	NA	NA	*	*	*	*	NA	*
lrg SFOV	120	260	4x5(4i)	1	4	16	±6.0	NA	±4.0	NA	NA	*	*	*	*	NA	*
																NA	*
35 Poly	120	400	4x5(4i)	1	4	16	NA	NA	±4.0	NA	NA	*	*	*	*		
	120	200	4x3.75(4i)	3	4	16	NA	<10	±3.0	NA	NA	*	*	*	*	NA	NA
	140	300	4x2.5(4i)	2	4	16	NA	<10	±3.0	NA	NA	*	*	*	*	NA	NA
	120	300	4x1.25(4i)	2	4	16	NA	<17	±3.0	NA	NA	*	*	*	*	NA	NA
35cm	120	300	4x2.5(4i)	2	4	16	± 5.0	<25	± 3.0	NA	NA	NA	NA	NA	NA	NA	NA

Note: NA - not applicable

* - see following page for detailed specification

Table 8-1 LightSpeed Ultra 1X Image Series Outline

GE PERFORMANCE PHANTOM

- **MTF**
 - Small Spot (< 200mA)
 - * Standard Algo.
 - * 3.8 lp/cm @ 50% Modulation
 - * 6.5 lp/cm @ 10% Modulation
 - Large Spot (> 200mA)
 - * Standard Algo.
 - * 3.8 lp/cm @ 50% Modulation
 - * 6.2 lp/cm @ 10% Modulation
- **MTF**
 - Small Spot (< 200mA)
 - * Edge Algo.
 - * 8.5 lp/cm @ 50% Modulation
 - * 13.0 lp/cm @ 10% Modulation
 - Large Spot (> 200mA)
 - * Edge Algo.
 - * 7.9 lp/cm @ 50% Modulation
 - * 12.1 lp/cm @ 10% Modulation

GE QA PHANTOM

MTF

- Standard Algo.
- **MTF** 0.65 - 1.0
- Contrast Scale 110 to 130

LCD	CONTRAST FACTOR	# OF HOLES VISIBLE
	2.0 to 3.99	at least 2
	4.0 to 7.99	at least 3
	8.0 to 12.0	at least 4

Table 8-2 GE QA Phantom - LCD

9 Detailed Calibration

If the 1X series Image Quality Tests fail then you need to perform DAS Gain Calibration and Complete Detailed Calibration. This would then need to be followed with another 1X Image Series.

- 1.) Click Left monitor **SCANNER UTILITIES** on bottom of the left monitor.
- 2.) Click **DAS GAIN CALIBRATION** from the displayed selections.
- 3.) Click **DETAILED CALIBRATION** from the displayed selections.
- 4.) Click **AUTO CT NUMBER ADJUST** from the displayed selections.
- 5.) Click **QUIT**.